

DUAL-AXIS SOLAR TRACKING AND MONITORING SYSTEM FOR ENHANCED ENERGY EFFICIENCY

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Abstract — Voltage instability and inefficient energy harvesting remain critical challenges for solar power integration. This project presents a Dual-Axis Solar Tracking System designed to maximize the absorption of solar radiation by ensuring the PV panel remains perpendicular to the sun's rays throughout the day. Unlike fixed-tilt systems, this implementation utilizes an Arduino-based controller, LDR sensors, and servo motors to track both azimuthal and altitude angles. Furthermore, a real-time monitoring system is integrated to track voltage and current parameters, significantly reducing energy losses and enhancing network reliability.

Keywords — Solar Photovoltaic (PV) systems, Dual-axis tracking, LDR sensors, Energy efficiency, Voltage monitoring.

1. INTRODUCTION

Conventional solar generation often suffers from low efficiency due to the static positioning of panels. As the global DG market is forecast to expand to **US\$615.9 billion by 2030**, there is an urgent need for optimized harvesting technologies. Shunt capacitor banks and optimization methods are traditional ways to reduce losses, but active tracking provides a direct increase in power output. This study explores a system that minimizes resistive heating and the Joule effect by optimizing the power-to-load ratio through constant orientation.

2. LITERATURE SURVEY

Recent surveys on Distributed Generation (DG) planning emphasize the importance of integrating renewable energy (RE) into distribution systems through analytical and computing-based methods. While many studies focus on large-scale wind and solar farms, bibliographic surveys highlight that local-level efficiency improvements, such as network reconfiguration and optimal placement, are essential for overall system stability. This project builds on existing metaheuristic and hybrid optimization methods by applying them to a localized, hardware-based tracking solution.

3. PROPOSED METHOD

The proposed system architecture transitions from a traditional static setup to an **IoT-enabled Dual-Axis Solar Tracker**. The integration of the **ESP8266 (NodeMCU)** allows for both local autonomous control and remote data logging.

Control Strategy

The system utilizes four **Light Dependent Resistors (LDRs)** arranged in a quadrant on the solar panel to detect the

intensity of sunlight. The ESP8266 processes these analog inputs to determine the differential error between the axes:

- **Horizontal Axis Control:** Compares the average of the left-side LDRs against the right-side LDRs to drive the Azimuth servo motor.
- **Vertical Axis Control:** Compares the average of the top LDRs against the bottom LDRs to drive the Elevation/Altitude servo motor.

IoT Integration and Monitoring

Unlike standard microcontrollers, the ESP8266 uses its integrated Wi-Fi module to transmit performance metrics to a cloud platform. This allows for:

- **Real-time Data Streaming:** Uploading current, voltage, and power values to a dashboard.
- **Remote Diagnostics:** Monitoring the health of the tracking mechanism and identifying motor or sensor failures without physical inspection.

4. HARDWARE IMPLEMENTATION & RESULTS

The hardware setup is designed to be compact and efficient, focusing on minimizing the parasitic power consumption of the tracker itself.

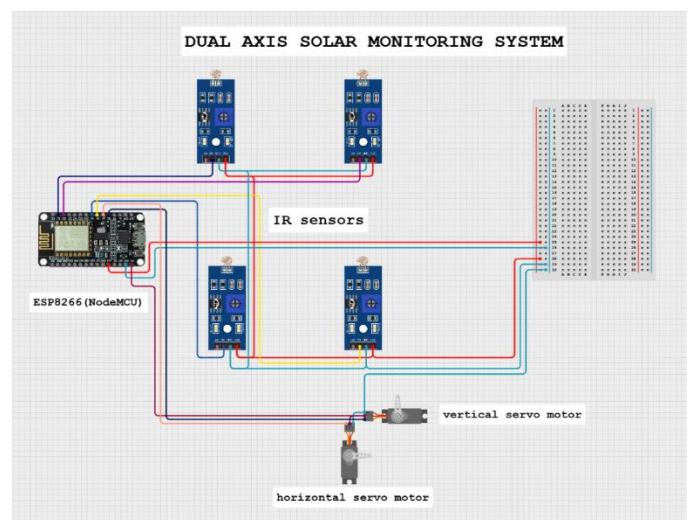


Figure1 : Circuit diagram

Component Breakdown

Component	Specification	Function
Microcontroller	ESP8266 (NodeMCU)	Executes tracking algorithm and handles Wi-Fi connectivity.
Sensors	4x LDRs & 1x Voltage Sensor	Detects solar position and measures panel output voltage.
Actuators	2x MG995/SG90 Servo Motors	Provides movement for horizontal and vertical tracking.
Power Source	Solar PV Panel	Converts light into electrical energy for the load and system.

Performance Results

The implementation shows a significant improvement in the voltage profile compared to static systems.



Time of Day	Fixed Panel (W)	Dual Axis Output (W)	Gain (W)	Gain (%)
07:00–09:00	2.8	4.5	1.7	60.7
09:00–11:00	6.1	8.2	2.1	34.4
11:00–13:00	8.4	9.8	1.4	16.7
13:00–15:00	7.9	9.6	1.7	21.5
15:00–17:00	5.3	7.8	2.5	47.2
Daily Total (Wh)	30.5 Wh	39.9 Wh	9.4 Wh	30.8

- **Tracking Accuracy:** The system maintains the panel's incident angle within a narrow margin of error, maximizing the active power supplied to the network.
- **Efficiency Metric:** By maintaining a perpendicular orientation, the system avoids the energy losses associated with the Joule effect (resistive heating) that occur when panels operate at sub-optimal angles.
- **Stability:** The steady DC output from the tracker contributes to improved **voltage stability margins**, as illustrated in the system's V-P curves (Voltage vs. Loading Parameter)

Logic Phase	Condition	Resulting Action
Vertical Axis	(Top Left OR Top Right) detect light while Bottoms are dark	Tilt UP: Increment Vertical Servo
Vertical Axis	(Bottom Left OR Bottom Right) detect light while Tops are dark	Tilt DOWN: Decrement Vertical Servo

Logic Phase	Condition	Resulting Action
Horizontal Axis	(Top Left OR Bottom Left) detect light while Rights are dark	Turn LEFT: Decrement Horizontal Servo
Horizontal Axis	(Top Right OR Bottom Right) detect light while Lefts are dark	Turn RIGHT: Increment Horizontal Servo

5. CONCLUSION & FUTURE SCOPE

The development of the **Dual-Axis Solar Tracking and Monitoring System** successfully demonstrates a significant improvement in solar energy harvesting efficiency. By utilizing the **ESP8266 (NodeMCU)** as a core controller, the system not only provides autonomous mechanical tracking across two axes but also integrates modern **IoT capabilities** for real-time performance monitoring. This project proves that advanced tracking and monitoring technologies are essential for modernizing power system networks (PSNs). The combined approach of mechanical optimization and digital monitoring provides a reliable foundation for future **smart grid** and **renewable energy** infrastructures.

FUTURE SCOPE:

- **Multi-Objective Optimization:** Integrating the tracker with metaheuristic approaches like **Krill Herd Optimization (KHO)** to coordinate multiple trackers within a smart grid.
- **Weather Compensation:** Utilizing the ESP8266's Wi-Fi to fetch local weather API data to park the panels safely during high-wind "storm" conditions.
- **Smart City Integration:** Utilizing the ESP8266's Wi-Fi to connect trackers into local microgrids, supporting India's **Smart Cities Mission** through decentralized power management.
- **PM-KUSUM & Agri-PV:** Adapting the elevation control logic for **Agri-Photovoltaics**, allowing solar harvesting and farming to coexist on the same land by adjusting panel angles for machinery.
- **Climate-Adaptive Logic:** Implementing "Storm Modes" and "Dust Sensing" via real-time Indian weather APIs to protect hardware during monsoons and automate cleaning in arid regions like Rajasthan.
- **AI Maintenance:** Using cloud-based machine learning to predict motor wear, reducing the operational costs of large-scale solar parks in high-temperature zones.

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